

Regional Integration and EU-ACP Trade

Correcting for a ratio regressor problem in gravity estimation

Noah Damski · 78 ACP countries × EU-27 + UK · 2000–2021 · 41,258 directed dyad-years

87%

OF THE NAIVE COEFFICIENT
VANISHES ONCE ONE
MECHANICAL CHANNEL IS
FROZEN

Abstract

A standard measure of regional integration divides a country's trade with its own bloc by its total trade, a denominator that includes trade with the EU, the very outcome the model explains. Because of that overlap, part of any coefficient on this measure is mechanical rather than economic. Holding the EU-trade share of the denominator fixed removes most of it: a large, seemingly robust stumbling-block coefficient (−2.287) falls by 87%, to a number no longer distinguishable from zero. Removing the ratio construction entirely, rather than partially, gives a cleaner test: entering its numerator and denominator as separate variables rejects the equal-and-opposite relationship a ratio assumes, in every version tested. Measured this way, the actual relationship is positive. Regional and EU trade expand together, not apart.

Reading the ratio

IT Share(i, t) =

intra-regional trade

intra-regional + other non-EU +

Σ_j X_{ijt}*

*Σ_j X_{ijt} is total trade with all 28 EU partners in year t, the same X_{ijt} from the equation to the right, summed across partners. It is also EU trade, the outcome the model explains.

So when EU trade rises for reasons unrelated to regional integration, IT Share falls anyway, before any behavior is involved. That is the arithmetic effect.

The problem sits in the denominator of IT Share itself. Total trade includes trade with the EU, so the regressor moves with the outcome by construction, separate from anything economic.

Estimating equation

X_{ijt} = exp(β ·ITShare_{it} + γ ·EPA_{it} + δ'Z_{ij} + μ_{jt} + η_i) ·ε_{ijt}

μ_{jt} EU-partner × year · η_i ACP country

Z: ln distance, common language, colonial tie

SEs clustered on ACP country (78). fixest, Bergé (2018).

X_{ijt} is the same bilateral trade flow on both sides of this poster: the outcome being explained here, and the quantity hidden inside IT Share's denominator on the left.

RESULTS

The baseline estimate

Coefficient on IT Share, PPML, 2000–2021

−2.287

SE 0.604, p < 0.001, n 41, 258

A one standard deviation rise in regional trade intensity is associated with **25.5% less** trade with the EU.

Robustness checks

CHECK	IT SHARE
Baseline PPML	−2.287 (0.604)
Negative binomial	−2.284 (0.531)
Sample from 1995	−2.555 (0.639)
Sample from 2007	−1.754 (0.519)
Four clustering schemes	SE 0.410–0.658
Drop any one bloc	−1.36 to −2.94

Every row stays negative and significant at 1%, except dropping ECOWAS alone, which falls to −1.356 (SE 0.839) and loses significance. ECOWAS's large mean IT Share makes it an outsized contributor to the pooled estimate. Using standard gravity alone, this is a confident stumbling-block result. None of these checks can detect a problem in what the variable itself measures, since the issue is in the regressor's construction, not in how the model is fit.

DIFFERENCES ACROSS BLOCS

Splitting this same baseline coefficient by regional bloc tests whether the effect is the same size everywhere. Four of the seven blocs are individually significant and negative: ECOWAS (−3.342), SADC (−3.439), CARIFORUM (−2.536), and PIF (−26.31, a thin bloc with wide standard errors throughout). Central Africa, the EAC, and ESA are not distinguishable from zero. A formal test confirms the blocs are not identical (F = 3.04, p = 0.010). Central Africa's and PIF's swings trace largely to currency arrangements: interacting IT Share with the CEMAC franc peg and Pacific dollarization shrinks each bloc's deviation from ECOWAS by 95% and 88%, respectively, to statistical zero. The EAC also deviates significantly from the pooled pattern but has no comparable currency explanation. Mean IT Share itself ranges from 0.043 (ESA) to 0.333 (SADC).

Holding the denominator fixed

Freeze the EU share of the denominator at each country's own long-run average, so the mechanical channel is switched off but the economics is untouched.

FLOW	BASELINE SHARE	EU COMPONENT FIXED
Total bilateral	−2.287 (.604)***	−0.289 (.661)
ACP exports to EU	−2.509 (.962)***	+1.212 (.969)
EU exports to ACP	−2.016 (.454)***	−1.554 (.429)***

The two regressors have similar means and standard deviations, so the columns compare directly. About 87% of the pooled coefficient goes with the denominator. The EU-exports row does not fall to zero here, which the restriction test to the right explains.

EXPORTS AND IMPORTS, SEPARATELY

Testing exports and imports on their own was part of the plan from the start, not an afterthought. The two baseline share coefficients above, −2.509 (ACP exports) and −2.016 (EU exports), are statistically indistinguishable from each other (difference −0.493, conservative SE 1.064).

Same story in logs

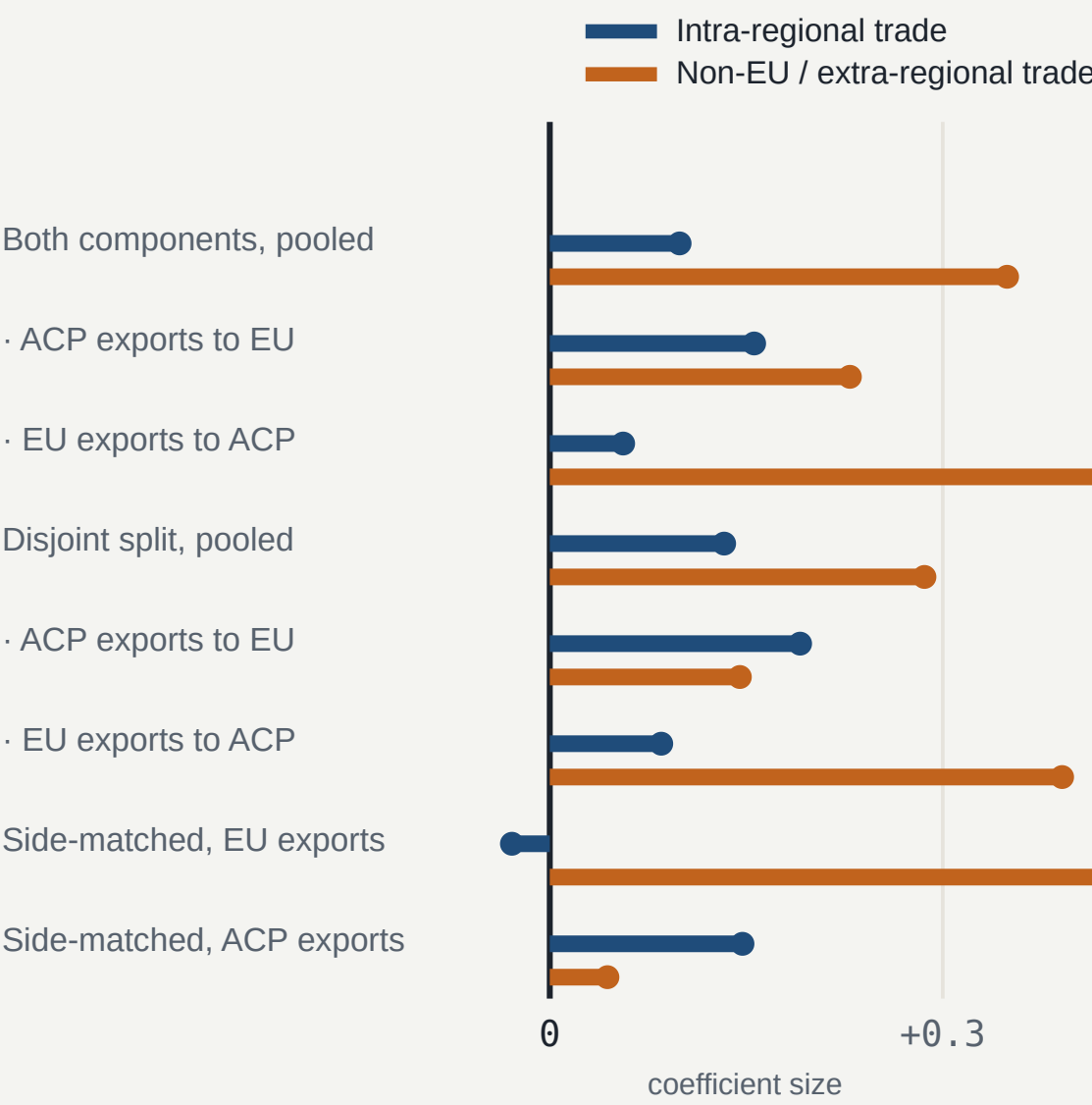
Holding the denominator fixed, above, asks whether removing the mechanical channel changes the coefficient. This asks a more basic question: is a ratio even the right way to combine intra-regional and non-EU trade in the first place? Enter both as separate regressors in the same PPML model instead of dividing them into one variable.

COMPONENT	COEFFICIENT
ln(intra-regional trade)	+0.099 (0.053)*
ln(non-EU trade)	+0.349 (0.064)***

Both come out positive instead of opposite in sign, exactly what a neutral ratio would require if it were dividing two unrelated things. This is the same regression as the first row of the restriction test to the right, run there in seven more versions.

Testing the ratio's restriction

A ratio forces a specific relationship. ln(intra-regional trade ÷ non-EU trade) is mathematically the same as ln(intra-regional trade) − ln(non-EU trade). So dividing the two into one variable silently assumes their coefficients are equal in size and opposite in sign, whether or not that is actually true. The test below checks it directly, in eight versions: **pooled** uses total non-EU trade as the second variable; **disjoint split** excludes intra-regional trade from that total so nothing overlaps; **side-matched** pairs import-side components with EU exports and export-side components with ACP exports.



Each pair is one regression, shown in the same units as the −2.287 estimate at far left. If the ratio's restriction held, blue and orange would point in opposite directions. In seven of eight pairs they don't, both point the same way. The one exception, "side-matched, EU exports," is explained below.

THE IMPORT-SIDE EXCEPTION

EU exports to ACP are the same flow as ACP imports, so this equation makes more sense measured against ACP's own import-side shares rather than its share of total trade. Even with that adjustment, the ratio version stayed negative: −1.180 (0.261), t = −4.53, significant. Entering import-side intra-regional and import-side non-EU trade as two separate variables instead resolves it: −0.029 (0.024), t = −1.20, not significant. The matching export-side test, for ACP exports to the EU, gives +0.147 (0.043), t = 3.39, positive. The negative reading was coming from the denominator here too, not from the underlying trade relationship.

What the corrected model shows

The "EU component fixed" test on the left still treats IT Share as a ratio: it only freezes one input to that ratio. This test removes the ratio completely. Intra-regional and extra-regional trade enter as two separate variables rather than one divided by the other, the same setup used for the restriction test. That is the full fix, which is why this result is clean and positive throughout under the log specification, while the EU-fixed numbers were mixed and one stayed significantly negative. Under an inverse-hyperbolic-sine transform that keeps zero-trade country-years instead of dropping them, this positive intra-REC result is not statistically distinguishable from zero, though the ratio's restriction is still rejected either way (Supporting Information, Table S.5).

+0.133

SE 0.053 · t = 2.49 · 95% CI [+0.028, +0.237]

Because both sides of this regression are in logs, this coefficient is an elasticity: it converts directly into a percent-for-percent relationship.

DIRECTION	ELASTICITY	T	+10%→
ACP exports to EU	+0.191 (.075)	2.55	+1.8%
EU exports to ACP	+0.085 (.042)	2.01	+0.8%
Extra-regional non-EU	+0.286 (.062)	4.65	+2.7%

TRADE AGREEMENTS (EPAS)

Economic Partnership Agreements, the EU's own trade deals with individual ACP countries or blocs, are a separate thing from the regional integration (IT Share) measured everywhere else on this poster. This check uses the original share specification, not the corrected model above. Recentering IT Share at its sample mean, the net EPA effect is −0.051 (SE 0.063, t = −0.81), not distinguishable from zero. Splitting by direction, though, EU exports to ACP fall roughly 12% under an in-force EPA (−0.126, SE 0.061), with no matching change on the ACP-exports side.

WHERE THIS PROJECT GOES NEXT

Instrumenting integration depth with tariff liberalization episodes or corridor infrastructure investment, to move toward a causal estimate. Extending the panel past 2021 to test whether the African Continental Free Trade Area changes this picture. Resolving the directional asymmetry left open above: intra-regional exports track ACP exports to the EU but not the reverse, and this paper does not yet explain why.